

Deep Learning - Theory and Practice

Basics of Machine Learning

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<http://leap.ee.iisc.ac.in/sriram/teaching/DL20/>

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Matrix Derivatives

$$\left(\frac{\partial \mathbf{a}}{\partial x}\right)_i = \frac{\partial a_i}{\partial x}$$

$$\left(\frac{\partial x}{\partial \mathbf{a}}\right)_i = \frac{\partial x}{\partial a_i}$$

$$\left(\frac{\partial \mathbf{a}}{\partial \mathbf{b}}\right)_{ij} = \frac{\partial a_i}{\partial b_j}.$$

Principal Component Analysis

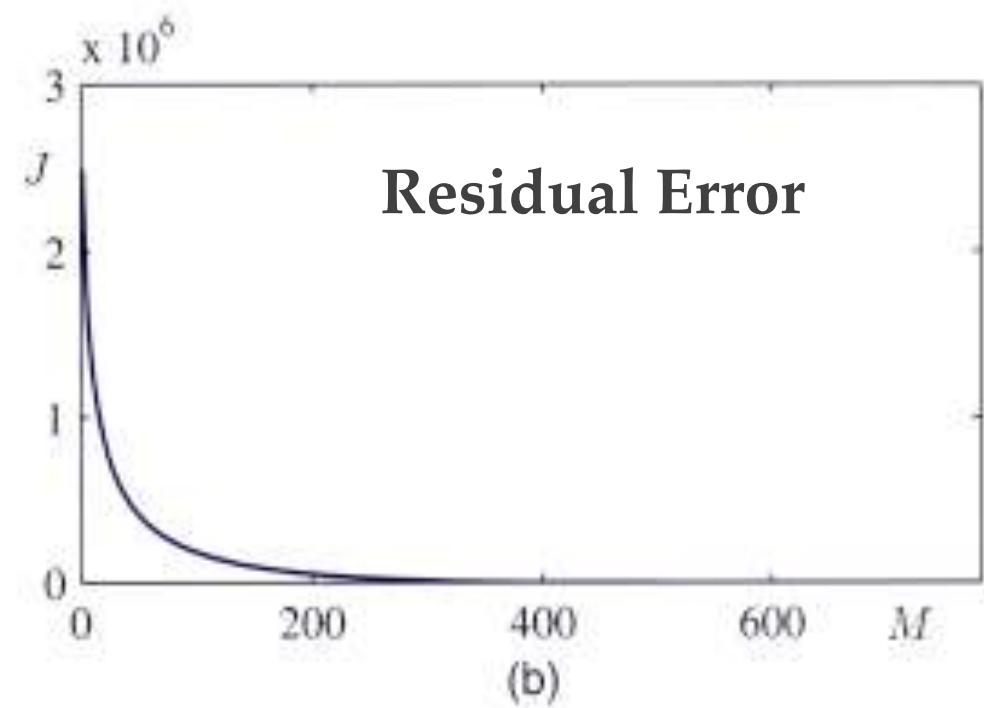
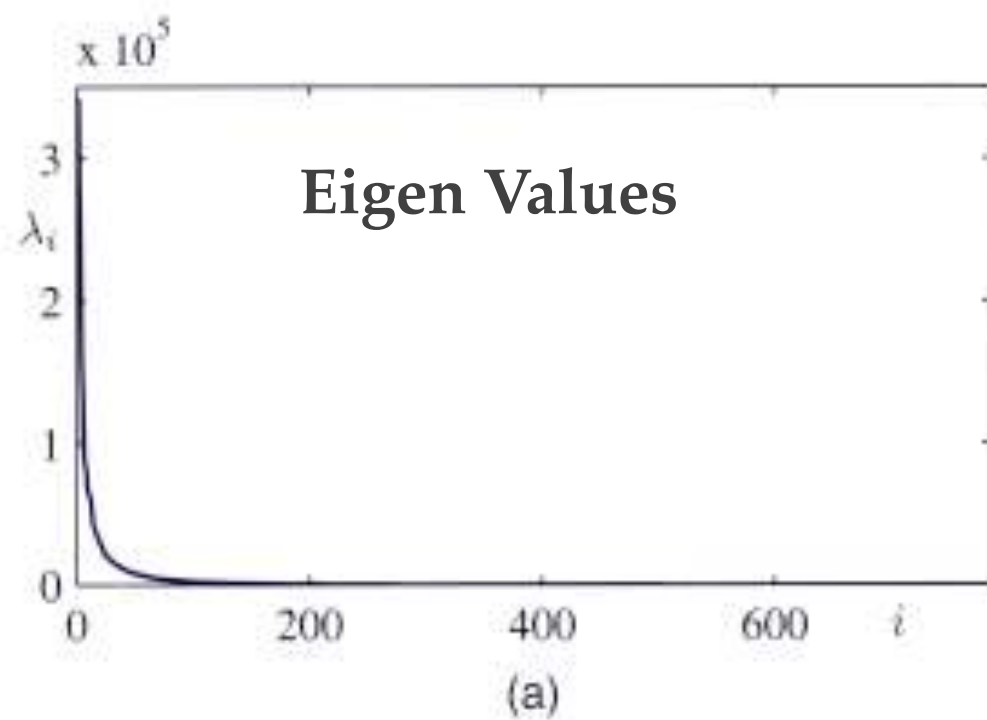
- ❖ First M eigenvectors of data covariance matrix

$$S = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^T$$

- ❖ Residual error from PCA

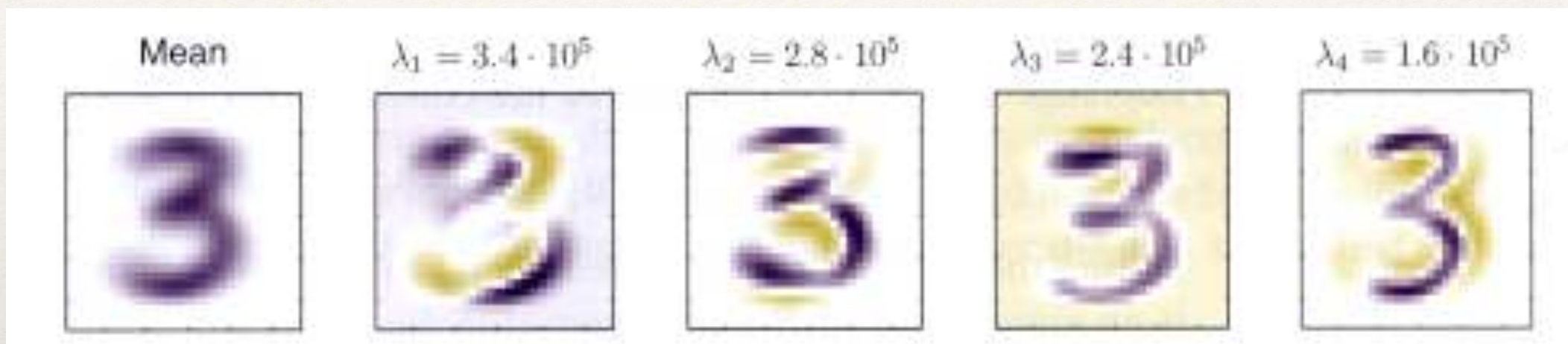
$$J = \sum_{i=M+1}^D \lambda_i$$

PCA

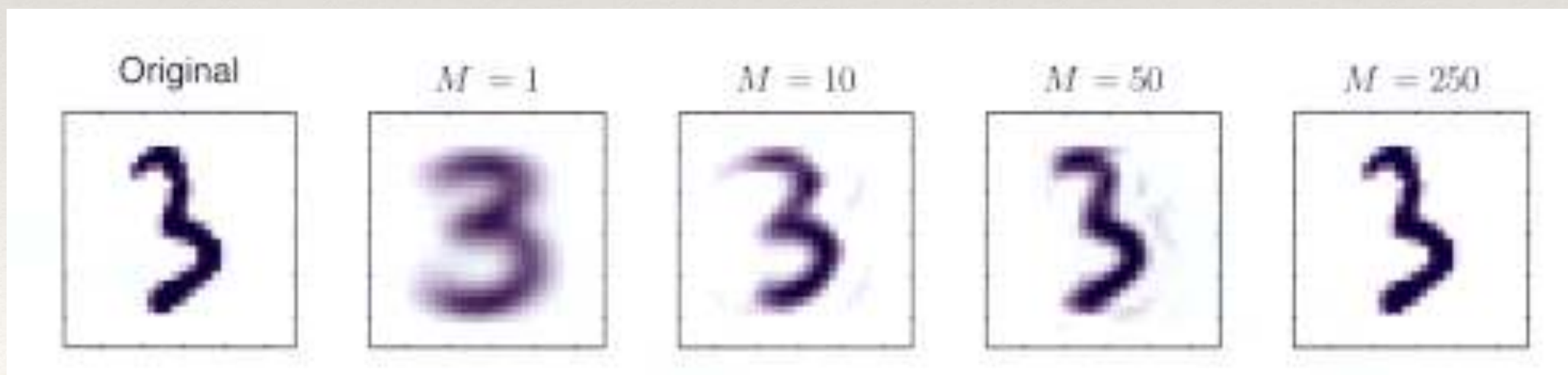


PCA - Reconstruction

Eigenvectors

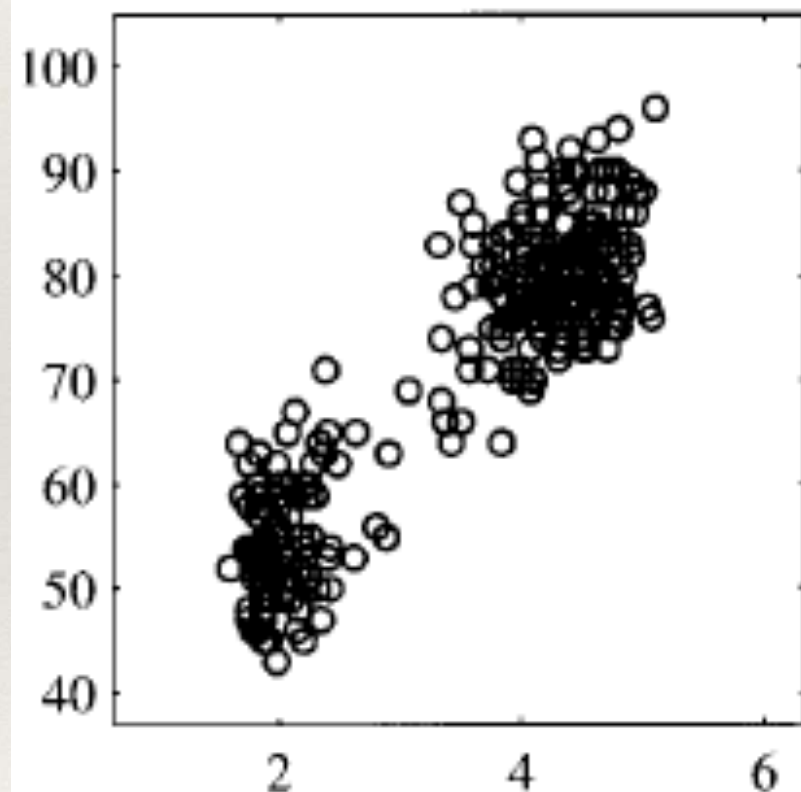


PCA - Reconstruction

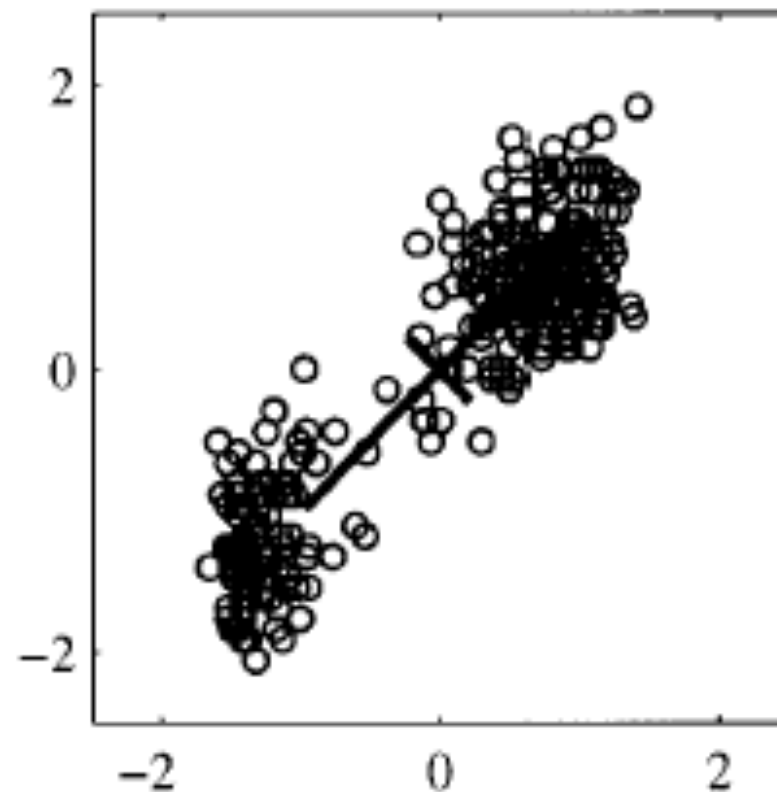


Whitening the Data

Original Data



Standardized Data



Whitened data

